Part 5

Professional Agentic Harness

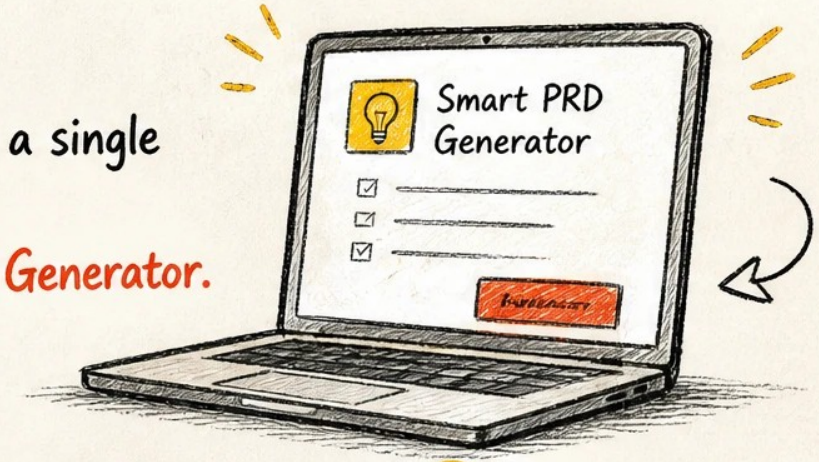


Ray Villalobos
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@planetoftheweb



FROM ONE PROMPT TO A LIVE PRODUCT

We started with a single agent prompt: the Smart PRD Generator.



1 CREATE

One prompt.
One clear idea.

2 SECURE

Protect users and data.

3 TEST

Make it reliable.
Make it right.

4 DEPLOY

Ship with confidence.
Go live.

5 LIVE

A real product.
Real users.



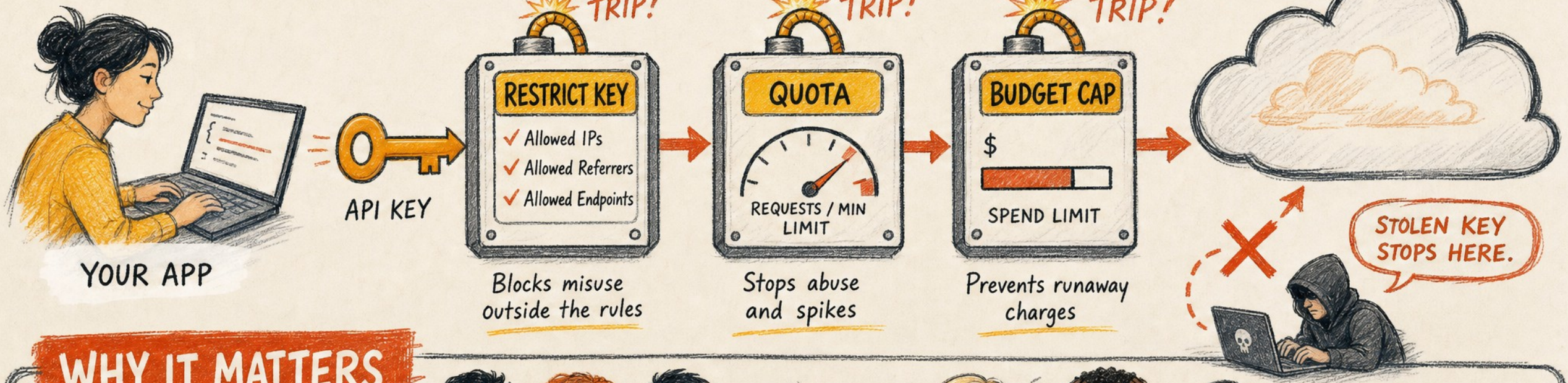
SEVEN DAYS
7

+ 68 COMMITS →

PAID PRODUCT AT MVPUNK.COM

THIS IS THE WHOLE ARC.
ONE IDEA. ONE WEEK. SHIPPED.

PUT A FUSE ON EVERY API KEY



WHY IT MATTERS

When a key leaks, the limits decide what happens next.



YOU LOSE $10.
Limits trip. Problem contained.

VS.



YOU LOSE EVERYTHING.
Runaway costs. Site goes down.

INTERNET STATUS: PATCH WEEK

PHP	Major outage
Node.js	Degraded
GitLab	Major outage
LiteLLM	Major outage
OpenCode	Degraded
eMudhra	Major outage
Your Product	Unknown



WHY IT MATTERS

Your product probably depends on one of these.



UPDATE



ROTATE



LIMIT







POLISH
THE OUTPUT
YOUR USERS HAND OFF



The thing your users paste into their AI is the product.

QUALITY IS A FEATURE.

BEFORE

- ✗ TBD – update this section
- ✗ We should add something here...
- ✗ Fix later, etc.
- ✗ The process is as follows...

```
function calc (x) {
  // TODO: handle edge cases
  return x * 2;
}
```

AFTER

- ✓ Calculate the total based on inputs.
- ✓ Add validation for required fields.
- ✓ Handle edge cases and errors.
- ✓ Process inputs and return result.

```
function calc(x) {
  return x * 2;
}
```

REMOVE FILLER → PRESERVE WHAT MATTERS → CLEAR, READY, RESPECTFUL → CONFIDENCE IN EVERY HANDOFF



- ✓ STRIPS FILLER WORDS (TBD, ETC.)
- ✓ LEAVES YOUR WORDS & CODE UNTOUCHED
- ✓ CLEAN, READY-TO-USE OUTPUT
- ✓ BETTER RESULTS, EVERY TIME

How we tested MVPunk

Tests stacked: Fast → Slow

FAST

SLOW

1. Unit checks

Run in milliseconds.

```
function add(a,b){
  return a+b;
}
```



2. Component tests

Render single pieces of the interface.



3. Smoke tests

Drive the whole app in a real browser.



★ Each layer runs at its cheapest gate.

1 Commit

Triggers unit tests.

```
git commit -m
"feat: add cart"
```



2 Push

Triggers smoke tests.



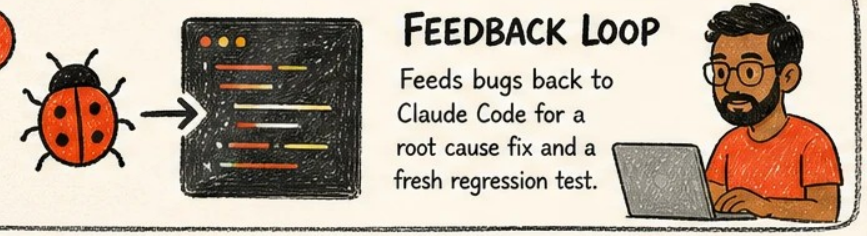
3 Browser agent

QAs the live site.



4 Feedback loop

Feeds bugs back to Claude Code for a root cause fix and a fresh regression test.



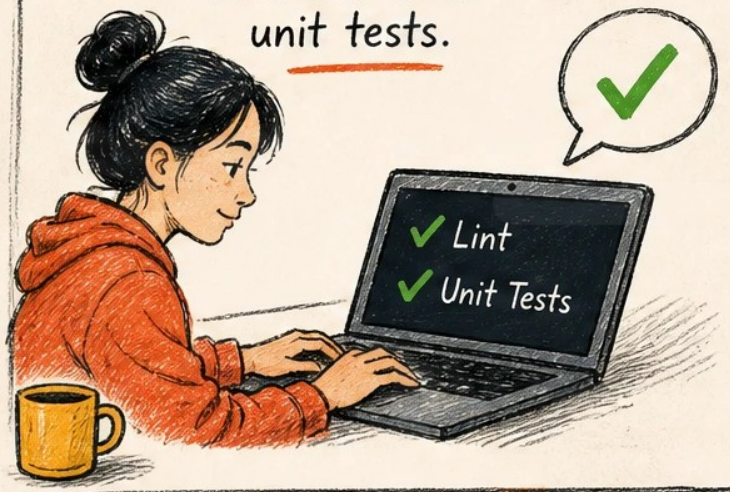
Commit Code → Unit tests Milliseconds → Component tests Interface Pieces → Smoke tests Real Browser → Browser agent QA Live Site → Bugs back to Claude Code → Fix + new regression test

AUTOMATE THE DISCIPLINE YOU'LL FORGET

Good habits automatic.

1 PRE-COMMIT HOOKS

Run lint and unit tests.



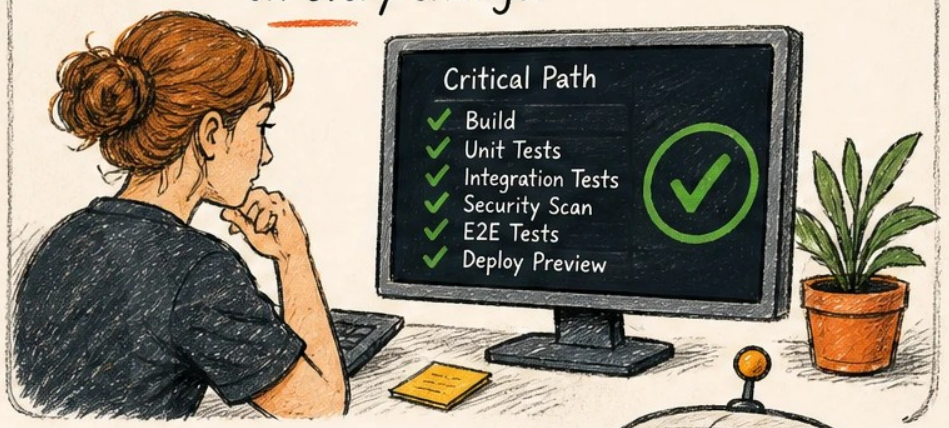
2 PRE-PUSH HOOKS

Run the smoke suite.



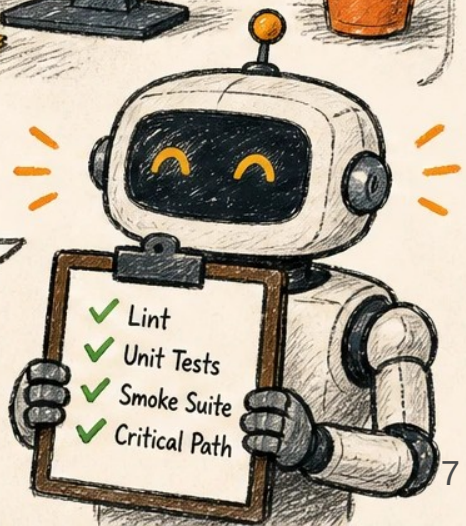
3 CI PIPELINE

Runs the full critical path on every change.

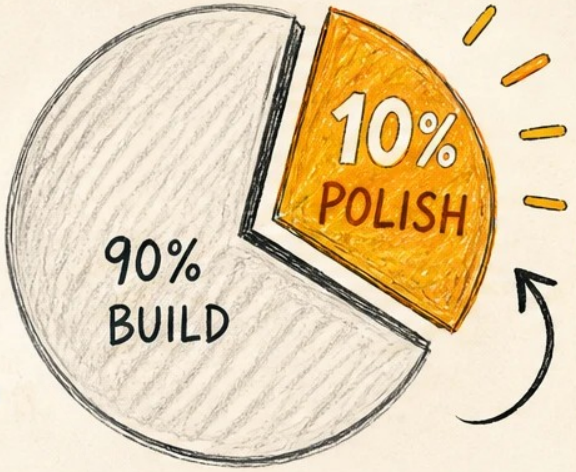




THE MACHINE REMEMBERS THE CHECKLIST SO YOU DON'T HAVE TO.



THE LAST TEN PERCENT IS WHAT LOOKS PROFESSIONAL



Small details.
Big difference.

This layer separates a demo from a product. Polish reads as trust.

1 A REAL DOMAIN



Custom domain.
Instant credibility.

2 SOCIAL CARDS



Look sharp everywhere you're shared.

3 FULL ICON SET



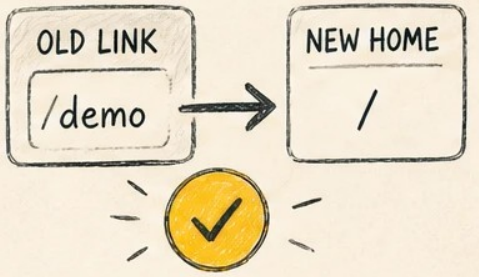
Consistent icons.
Clear experience.

4 WEB APP MANIFEST



Works like an app.
Built-in trust.

5 OLD LINKS KEEP WORKING



No broken links.
No lost trust.



Polish reads as trust.

START AT THE COST FLOOR



Your first number
is your variable cost per unit.

Each MVPunk generation costs about **$0.55** in Gemini API calls.

That figure is **the floor** and every plan on the menu has to clear it.

You **lock the floor** before you ever glance at a competitor's price tag.

$0.55
PER MVPUNK GENERATION
(GEMINI API CALLS)

COST FLOOR: $0.55

Know Your Numbers.



HOW WE PRICED MVPunk

PRICING STARTS AT THE COST FLOOR.



Each generation costs about **$0.55** in model calls and no plan may dip below it.

TRANSLATE EVERY PLAN INTO DOLLARS PER CREDIT TO FIND THE LOSS LEADER.

PLAN	$/CREDIT
Starter	$0.55
Pro	$0.57
Max	$0.58
Pay As You Go	$0.70
Mini Pack	$1.20



The loss leader sets the entry point for growth.



SUBSCRIPTION RUNS NEAR BREAK EVEN...



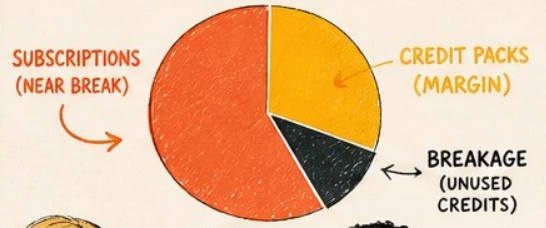
...while small credit packs carry the margin.



MONTHLY CREDITS ROLL OVER TO A CAP AND UNUSED ONES EXPIRE.



BREAKAGE AND MIX KEEP THE PLAN SELF SUSTAINING.





Fair. Simple. Predictable.

★ *Smart pricing. Happy users. A business that lasts.*

GITHUB

Code. Collaborate. Ship.

GitHub is the world's top platform for building software together.

- HOST YOUR CODE
- COLLABORATE ANYWHERE
- SHIP BETTER SOFTWARE

HOW DEVELOPERS USE GITHUB

CODE MANAGEMENT



Store, organize, and version your code with Git.

COLLABORATE WITH TEAMS



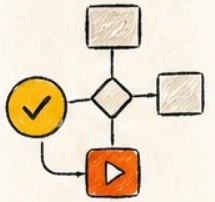
Branch, open Pull Requests, review and discuss changes.

REVIEW & MERGE WITH CONFIDENCE



Catch issues early with code reviews and automated checks.

AUTOMATE WORKFLOWS



Use GitHub Actions to test, build and deploy automatically.

MANAGE PROJECTS



Track tasks, bugs and milestones in one place.

VIBE CODERS LOVE GITHUB ♡



START ANYWHERE
Spin up ideas, save code, and access it anywhere.

FORK & EXPLORE
Fork projects, remix, and learn by doing.

GET HELP & SHARE
Join a massive community, ask questions, and share wins.

SHOWCASE YOUR WORK
Your portfolio lives here. Build in public.



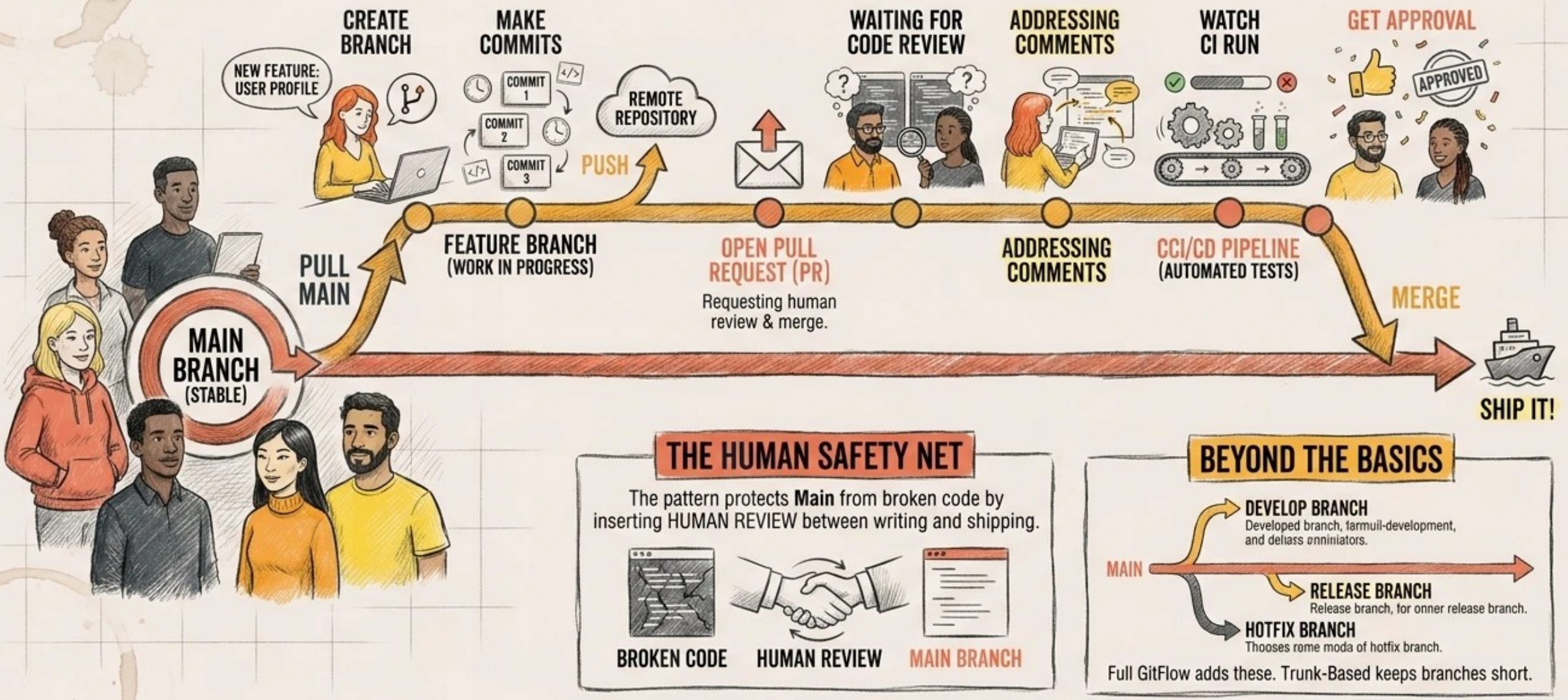
IN ONE PLACE

- ✓ Code
- ✓ Collaboration
- ✓ Automation
- ✓ Community

From first commit to global impact—GitHub is where developers and vibe coders build, connect, and create the future.

Ship with confidence.

TRADITIONAL GITFLOW: HUMAN-DRIVEN DEVELOPMENT DANCE



THE HUMAN SAFETY NET

The pattern protects **Main** from broken code by inserting HUMAN REVIEW between writing and shipping.

BROKEN CODE | HUMAN REVIEW | MAIN BRANCH

BEYOND THE BASICS

Full GitFlow adds these. Trunk-Based keeps branches short.

THE SHAPE STAYS THE SAME: BRANCH, BUILD, REVIEW, MERGE. Choreographed & Human-Driven.

AI in GitFlow

AI: I'll draft the feature code and write conventional commits.

AI: I've generated the PR description from the diff for you.

AI: Running automated reviews...

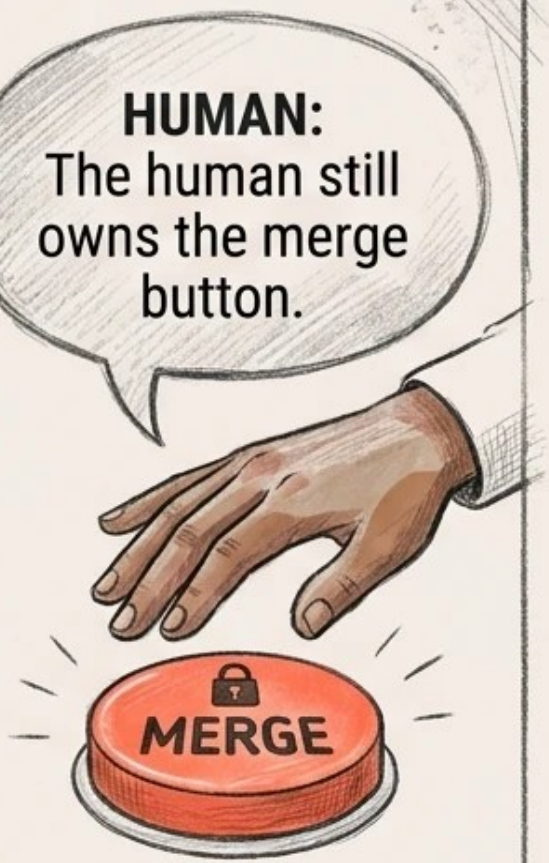
AI: CI logs read. Tests failed!

AI: Proposing fixes for the failed tests.

AI: Walking you through merge conflicts and nudging on old PRs.

HUMAN: The human still owns the merge button.





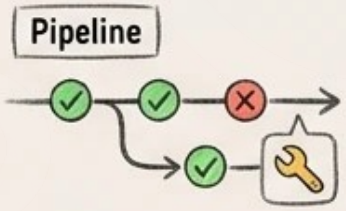
FEATURE BRANCH → **PULL REQUEST (PR)** → **AUTOMATED CHECKS & CI** → **CODE REVIEW & CONFLICTS** → **FINAL MERGE APPROVAL**

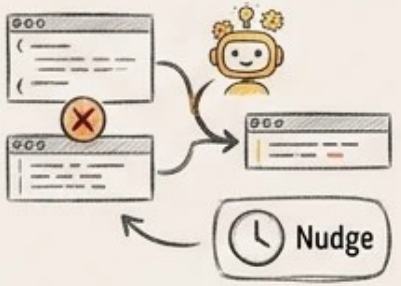
- feat: new login module
- fix: handle edge cases

PR description

- Summary of changes...
- Key features added...
- Addressing issue #123

Pipeline





SAME CHOREOGRAPHY. AI SHOULDER-TAP AT EVERY STEP.

THE REPO'S PAPERWORK

Every professional repository ships with non-code files that quietly run the show.
Boring files that signal you take the project seriously.

README





Explains how to install, run, and contribute, the front door for any stranger.

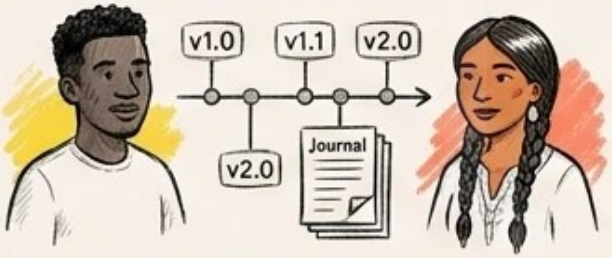
LICENSE





Makes the code legally usable, with MIT for permissive use, Apache 2.0 for patent protection, and GPL for viral copyleft.

CHANGELOG



Records human-readable changes per version.

.gitignore



Keeps secrets and build artifacts out of source control.

CONTRIBUTING & CODE_OF_CONDUCT





Set the rules for participation.

YOUR KEYS HIDE IN THREE PLACES



1 GIT HISTORY

Old commit still holds a deleted key

```
3f2a1b7  feat: add payments
a1c9d3e  refactor
b7e9c2d  (deleted)
API_KEY=sk_live_51H8...
d4e6f8a  fix: typo
main
```

Still here.
Still risky.

2 BROWSER BUNDLE

Key shipped to every visitor

```
// bundle.js
const config = {
  apiKey: "pk_live_a1B2..."
}
```

Delivered to every visitor.

3 .ENV FILES

Secrets sitting on servers & laptops

```
.env
DB_URL=postgres://...
API_KEY=sk_live_6F29...
STRIPE_SECRET=whsec_...
EMAIL_PASS=supersecret
...
```

Easy to leak.
Hard to detect.

WHY IT MATTERS

Deleting a key from code is not enough.

SCAN HISTORY

Search your git history and remove exposed secrets.

KEEP SECRETS ON THE SERVER

Never ship real keys to the browser.

ROTATE LEAKED KEYS

Treat any exposed key as compromised.

Assume it's out. Act like it is.

THE PRIVACY CHECK

1 Read the repo



2 Ask 0 to 3 questions



3 Check





Policy matches the code



Consent first



Collect only what you need



See, export, delete



Where the data goes



Kids under 13



WHY IT MATTERS

Users trust apps that are honest about their data.



Educational checks, not legal advice.

THE TOOL LANDSCAPE: PRO AI DEVELOPMENT TOOLS

AGENT-AWARE IDES:
TEXT EDITORS WITH SUPERPOWERS





- CURSOR
- CLAUDE CODE IN IDE
- CODEX IDE
- OPEN SOURCE OPTIONS: KILO CODE, OPENCODE

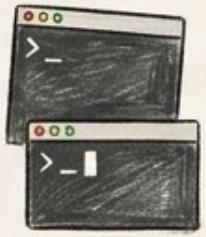
BOOSTED PRODUCTIVITY & CREATIVITY

AGENTIC CLIS:
COWORKERS WHO LIVE IN YOUR SHELL





- CLAUDE CODE CLI
- CODEX CLI
- GEMINI CLI
- GITHUB CLI

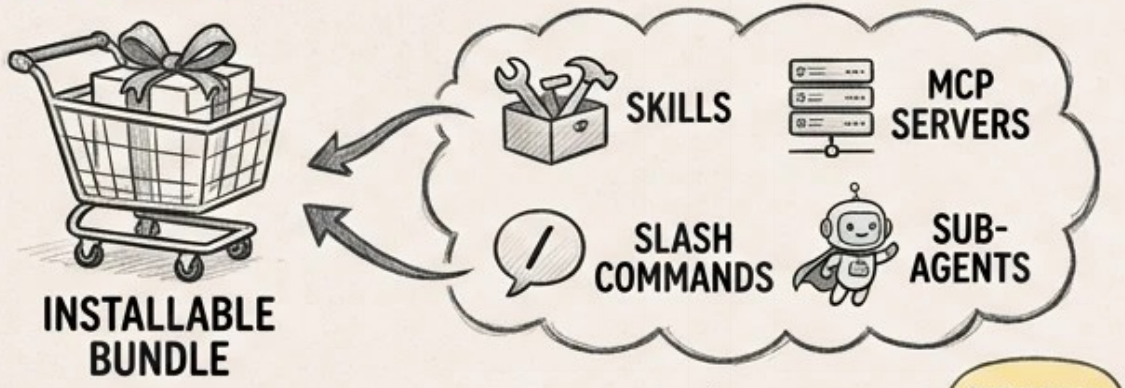


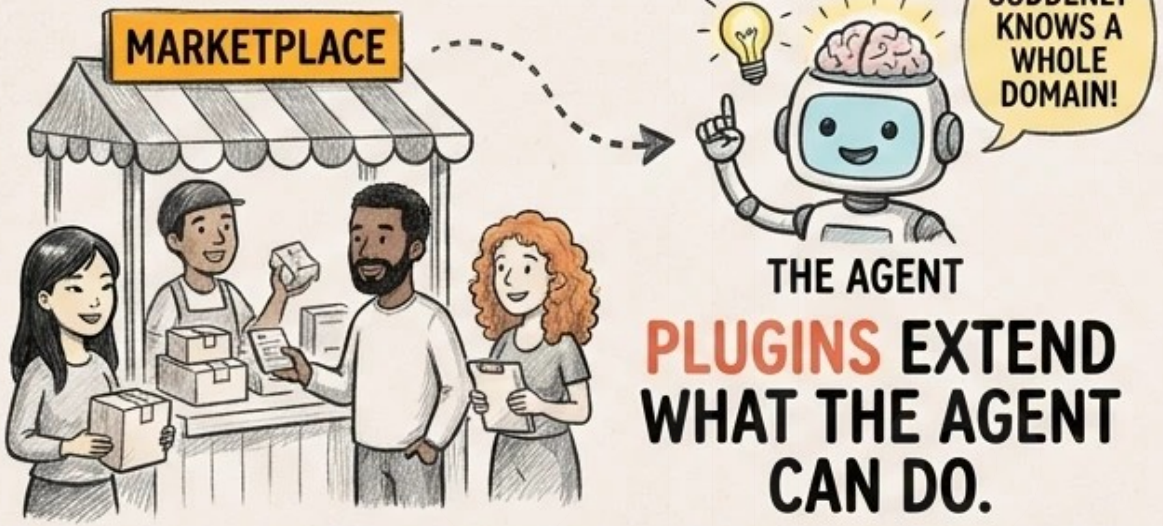
SEAMLESS COLLABORATION & AUTOMATION

THE EVOLVING WORKFLOW: CHOOSE YOUR AI DEVELOPMENT PARTNER

PLUGINS AND HOOKS: TWO WAYS TO CUSTOMIZE AGENT BEHAVIOR

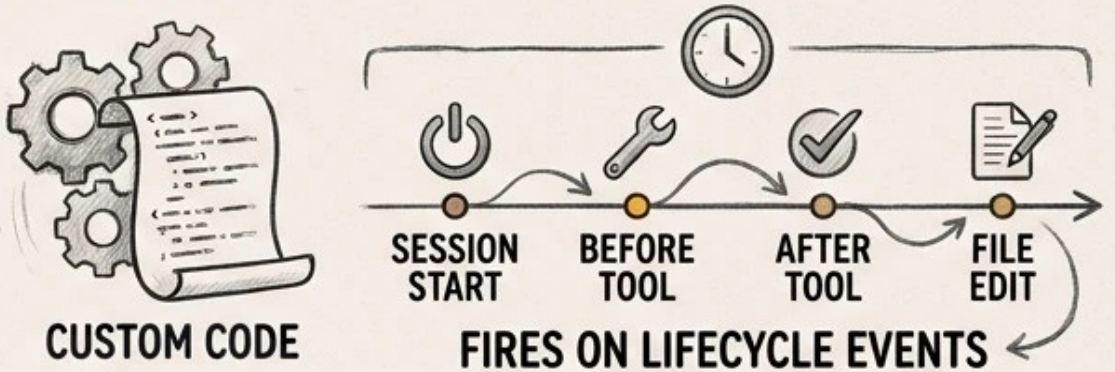
PLUGINS: INSTALLABLE BUNDLES



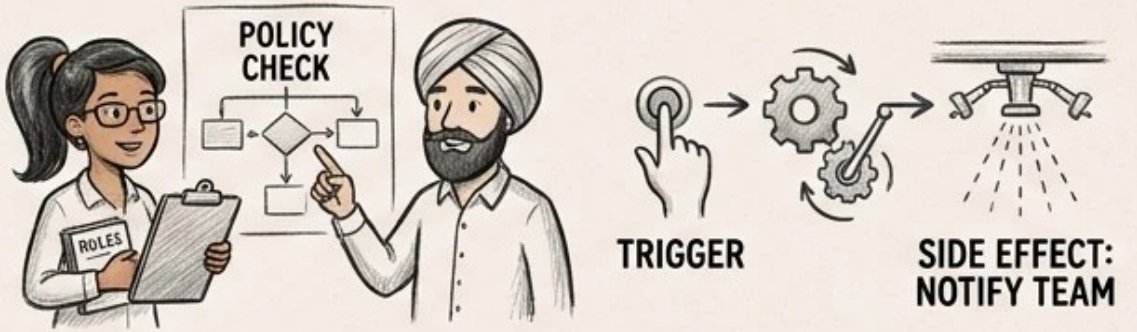


PLUGINS EXTEND WHAT THE AGENT CAN DO.

HOOKS: CUSTOM CODE ON EVENTS



ENFORCE POLICY OR AUTOMATE SIDE EFFECTS



HOOKS CONTROL HOW IT DOES IT.

CUSTOMIZE YOUR AGENT: EXTEND CAPABILITIES & CONTROL EXECUTION!

SUBAGENTS

MAIN AGENT
DISPATCHES WORK &
ASSEMBLES ANSWER



EXPLORER:
Reads the codebase



PLANNER:
Drafts the approach



REVIEWER:
Checks the output

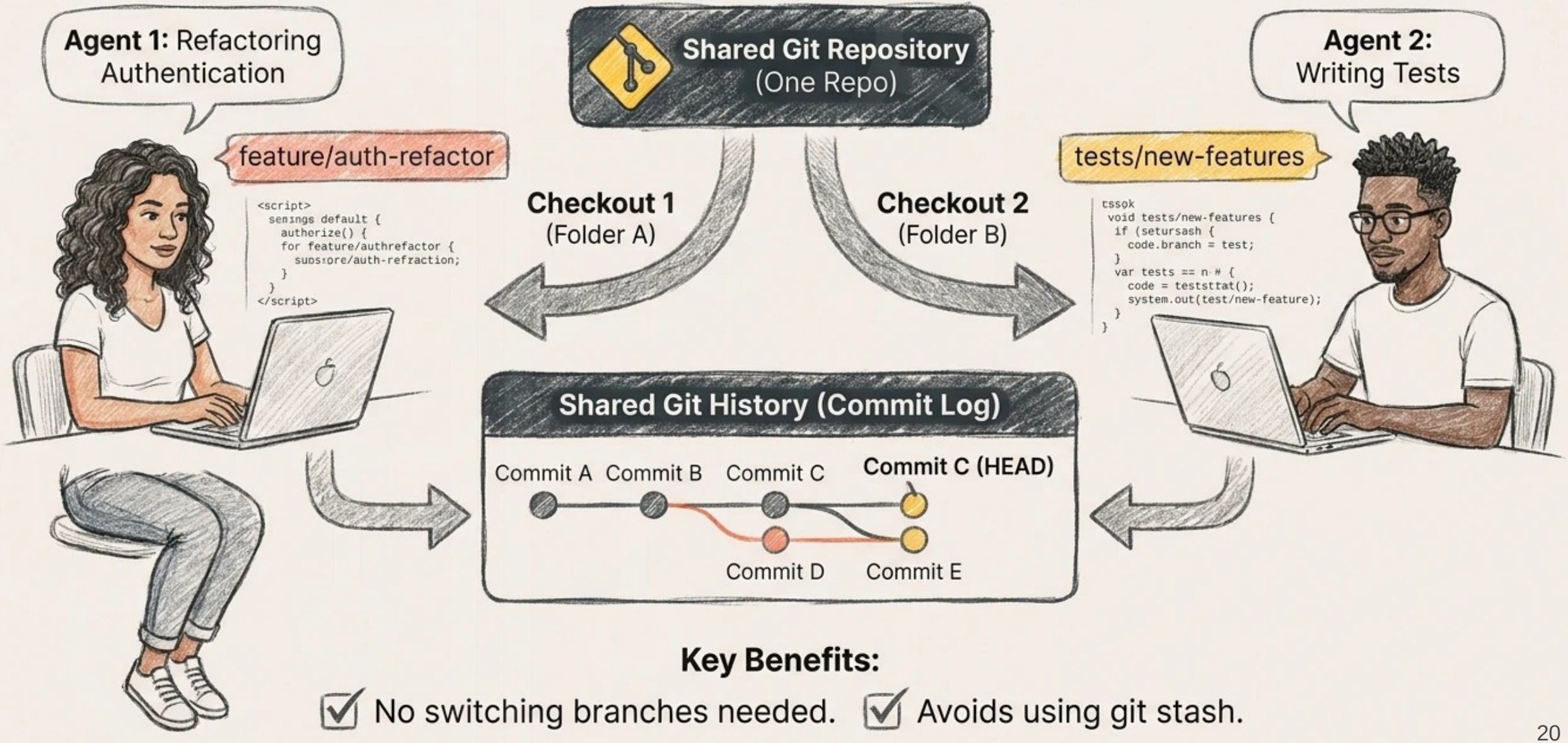


TESTER:
Runs the test suite

Each subagent runs in its own context window.
Reasoning doesn't cross-pollute.
Parallel work, specialized expertise, single answer.

FINAL ASSEMBLED ANSWER

GIT WORKTREES: Parallel Work in One Repo

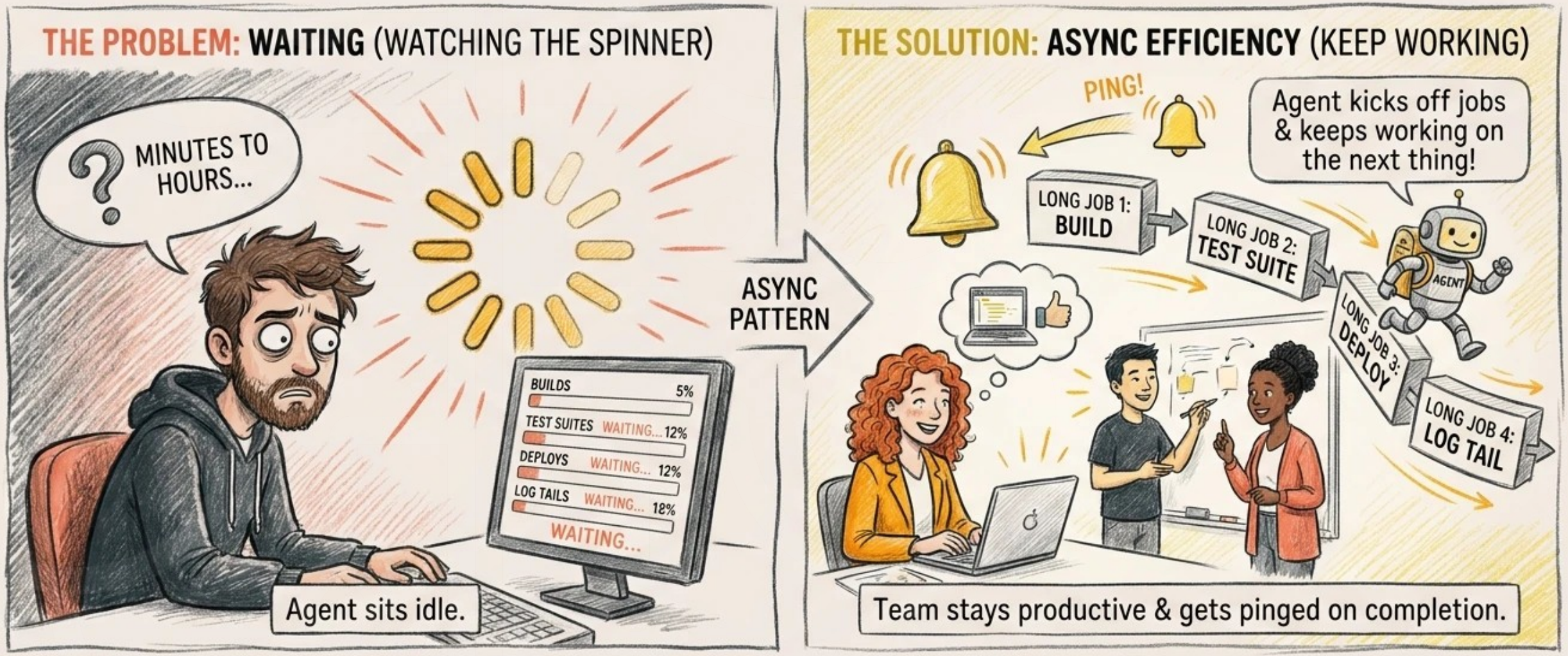


Key Benefits:

☑ No switching branches needed. ☑ Avoids using git stash.

DON'T WATCH THE SPINNER. USE BACKGROUND TASKS.

Let the agent run long jobs while you keep working.



With background tasks, the agent never sits idle watching a spinner. Enjoy the flow.

CLOUD AGENTS

Remote, Sandboxed Autonomous Workers.

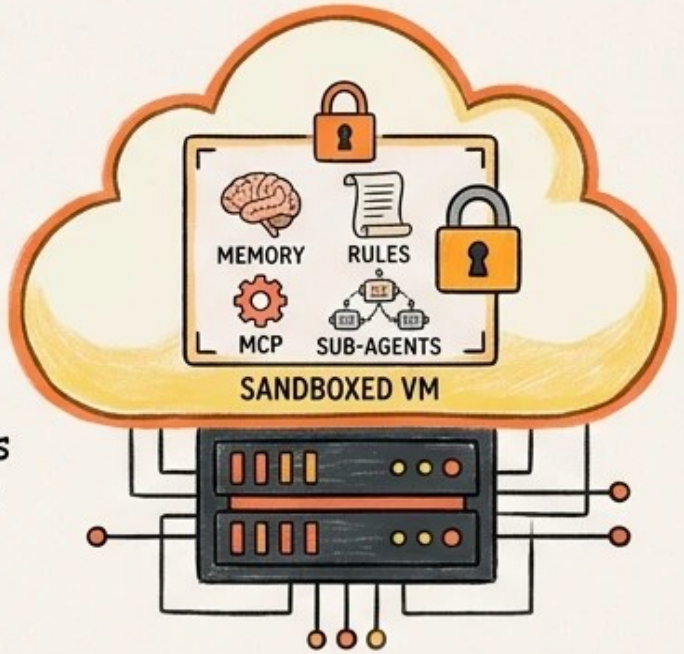


LOCAL AGENT
(Iteration & Dev)

Runs on your machine. Great for quick changes & testing, but ties up your resources.

CLOUD AGENT
(Long Autonomous Runs)

Runs in a secure remote VM. Keeps working overnight on complex tasks without using your laptop.



UNINTERRUPTED WORKFLOW



Perfect for long refactors, migrations, and high-token jobs.

CONTAINED BLAST RADIUS



Safest place to run untrusted code. Isolates risks from your local system.

EXAMPLES & USE CASES

- Anthropic Cloud Agents
- Codex Cloud
- Devin



LOCAL for ITERATION, CLOUD for LONG AUTONOMOUS RUNS.

/LOOP, /GOAL, /SCHEDULE

IN CLAUDE & CLAUDE CODE

/LOOP

Repeat a task, refine, and keep improving.



Great for iterative work and continuous progress.

/GOAL

Define the destination. Claude helps plan and get there.



Aligns work with what really matters.

/SCHEDULE

Set when things should happen. Claude keeps it on track.



Time it right. Stay consistent.

Different tools. One partner. You set the direction. Claude helps you get there.

LOOPS NEED A REAL GATE

REAL GATES

- Failing test
- Compiler error
- Static analyzer
- Reviewer reads the diff

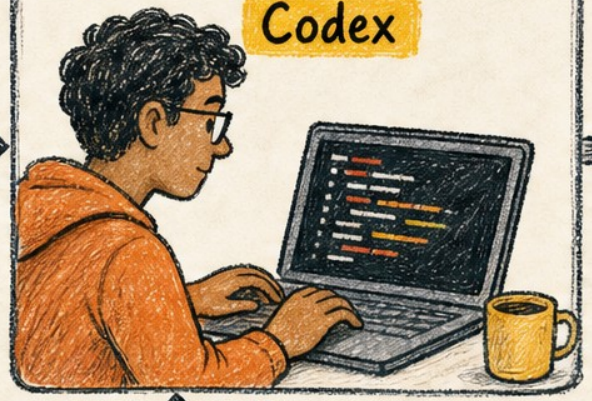
1 INVESTIGATE

Gemini



2 BUILD

Codex



3 CHECK

Grok





MAX 3 ROUNDS

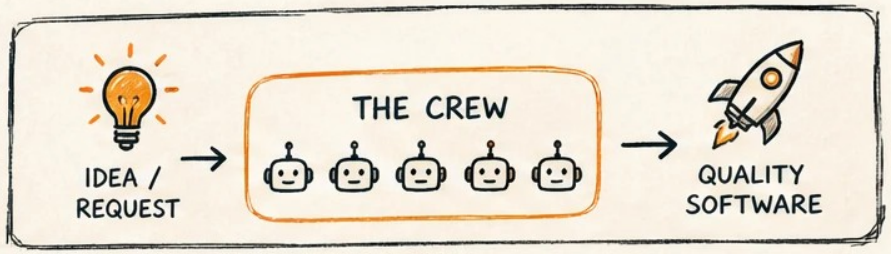
WHY IT MATTERS:

An agent grading its own work drifts.
A separate checker catches what the builder missed.



YOUR AI DEVELOPER TEAM

Built with CrewAI to plan, build, and ship better software—together.

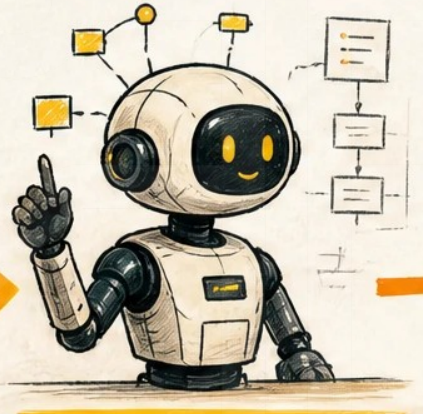




1. CREW MANAGER

Sets goals, defines tasks, manages priorities and keeps the team aligned.

- Define Objectives
- Assign & Track Tasks
- Monitor Progress



2. ORCHESTRATOR / PLANNER

Breaks down goals into steps, plans the work, and coordinates the team.

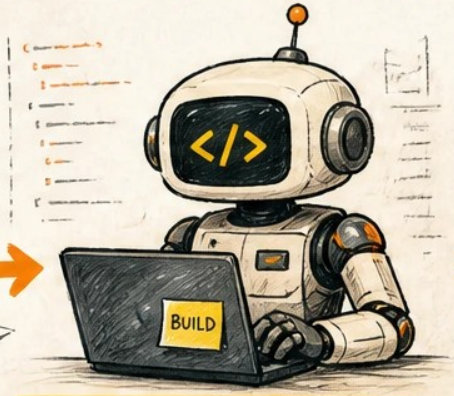
- Decompose & Plan
- Route to Right Agent
- Adapt & Re-Prioritize



3. DESIGNER / ARCHITECT

Designs solutions, structures systems and defines interfaces & data flows.

- System & UX Design
- Architecture & Patterns
- Specs & Documentation



4. ENGINEER / IMPLEMENTER

Writes code, builds features, and integrates everything end-to-end.

- Code & Implement
- Integrate & Test
- Optimize & Refactor



5. REVIEWER / VERIFIER

Reviews code and designs, finds issues, and ensures quality & compliance.

- Code Review & QA
- Find Issues & Risks
- Validate & Approve

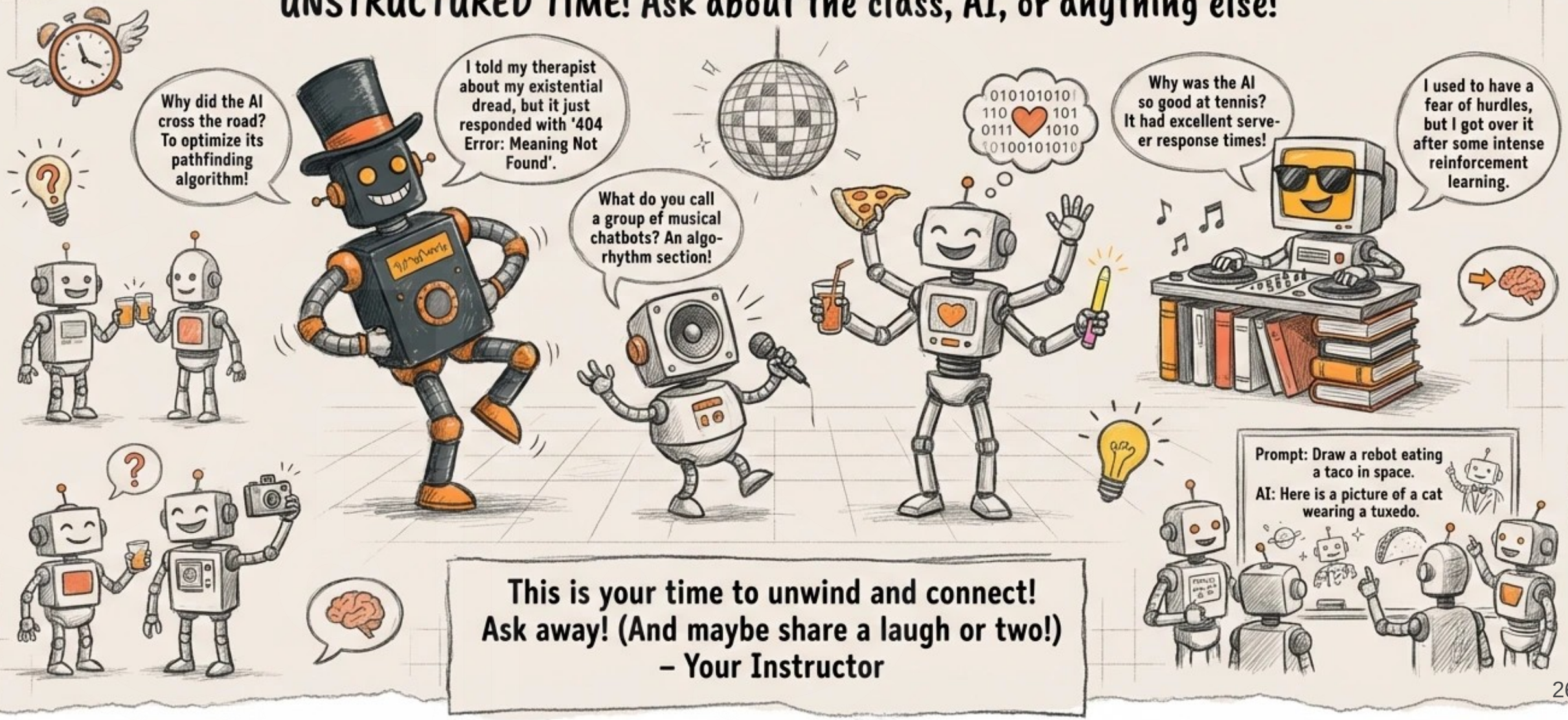
POWERED BY THE RIGHT MODELS

Fast & Smart (Reasoning)	Creative & Deep (Design)	Strong & Precise (Coding)	Critical & Accurate (Review)
gpt-4o / claude-3.5 for planning & coordination	claude-3.5 / gemini 1.5 for architecture & docs	gpt-4o / deepseek-coder for implementation	claude-3.5 / o3-mini for reviews & testing

CrewAI orchestrates the team, tools, and knowledge—so your ideas become reliable, high-quality software.

AMA (ASK ME ANYTHING) AFTERPARTY: THE CLASS CONCLUSION

UNSTRUCTURED TIME! Ask about the class, AI, or anything else!



This is your time to unwind and connect!
Ask away! (And maybe share a laugh or two!)
– Your Instructor